

Course Plan

Introduction to Programming

General Information

Degree Program: Computer Science / Software Engineering
Level: First-year undergraduate
Duration: 16 weeks
Format: Theory and Practice

Course Description

This course introduces the fundamental concepts of computer programming. Students will develop logical thinking and problem-solving skills through the use of algorithms, pseudocode, and a high-level programming language.

Course Objectives

- Understand the basic concepts of programming.
- Develop algorithms to solve computational problems.
- Apply control structures and basic data types.
- Introduce good programming practices and code documentation.

Course Topics

- Introduction to programming and algorithms
- Variables, data types, and operators
- Control structures (conditionals and loops)
- Functions and procedures
- Arrays and basic data structures
- Introduction to modular programming
- Basic error handling
- Good programming practices and code documentation

Teaching Methodology

The course will be conducted through hands-on sessions, problem-solving exercises, case studies, and small programming projects.

Assessment

- Practical assignments and exercises: 40%
- Midterm exams: 40%
- Final project: 20%

Prerequisites

Basic computer skills. No prior programming experience is required.